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12 May 1975
SUPERSEDING
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5 March 1962

MILITARY SPECIFICATION

CARTRIDGE, 90MM, HEAT-T, M431A2 LOADING, ASSEMBLING, AND PACKING

This specification is apprcved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive anti-tank cartridge designated as Cartridge, 90MM, HEAT-T, M431A2, for use in 90MM guns M36, M41 and M54.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-S-13812	- Steel Plate, Wrought Homogeneous, for Ammunition Testing (1/2 to 12 inches inclusive)

FSC: 1315

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributies (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	- Inspection, Radiographic
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8822481	- Cartridge, 90MM, HEAT-T, M431A2, Assembly
8800077	- Box, Packing, Ammunition, Wirebound, for Cartridge, 90MM, HEAT-T, M431, M431E1 and M431A2 for 90MM Gun, M36, M54, M41 in Fiber Container M411
8800078	- Container, Ammo, Fiber, M411, for Cartridge, 90MM, HEAT-T, M431, M431E1 and M431A2 for Gun M36, M54, M41
7258441	- Gage, Profile and Alignment
7258442	- Gage, Alignment Check
7258443	- Gage, Wear Limit Check

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material

3.1.1 General.-Materials shall be in accordance with applicable drawings and specifications.

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3.1.2 Clean scrap.-Clean scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.5).

3.2 Cartridges.-The cartridges shall comply with all requirements specified on Drawing (dwg.) 8822481 and with all requirements specified in applicable specifications.

3.3 Cavitation and Explosive Leakage

3.3.1 General.-Cavities having a maximum (max.) projected dimension of 1/32 inch or less shall be disregarded provided that they are separated by 1/2 inch of solid explosive.

3.3.2 Maximum projected area and dimensions of cavities.-The max. projected areas and dimensions of cavities in each of the segments (see Figure 1) shall be in accordance with Table 1.

TABLE 1

SEGMENT A		SEGMENT B	
Sum of areas of all cavities (square inches)	Maximum Dimension of any cavity (inch)	Sum of areas of all cavities (square inches)	Maximum dimension of any cavity (inch)
1/16	3/8	3/16	1/2

SEGMENT C
Sum of areas of all cavities (square inches)
1/4

3.3.3 Cavities extending into two segments.-The projected area of cavities extending into two segments shall meet the requirement of both segments.

3.3.4 Piping Cavity.-A piping cavity on the longitudinal axis of the projectile between the apex of the cone and the

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bottom surface of the fuze cavity shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/8 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.3.5 Surface cavitation.-Cavities formed due to explosive chipped or broken from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The maximum dimension of any visible cavity, after machining, on any surface of the fuze well shall not exceed 1/8 inch.

a. The total area of all cavities visible on the bottom surface of the fuze well shall not exceed 1/4 square inch. (Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation.)

b. A cavity, surrounding the conduit and extending into the fuze well, formed by crumbling explosive will be permitted. Size of this cavity is limited to 1/4 inch width, 1/4 inch radially and 1/2 inch in length measured longitudinally from the base of the wire slot in the body. An additional cavity in the area of the unused wire slot, limited to 1/4 inch width, 1/8 inch radially and 1/4 inch in length measured longitudinally from the base of the unused wire slot and extending into the fuze well, will also be permitted.

c. Crumbling of explosive around the rim of the fuze well, limited to 1/16 inch (max.) X 1/16 inch (max.) X 360° is permitted and may be disregarded in cavitation computations.

3.3.6 Cracks and annular rings.-Cracks shall be permitted if they are no wider than 1/32 inch. The presence of annular rings 1/16 inch thick maximum, and extending from the shell wall 1/4 inch into the charge shall not be cause for rejection, except for Segment C. No annular rings are permissible in Segment C. The cavity area represented by such a crack or annular ring shall not be included when determining cavity area in the charge.

3.3.7 Leakage of explosive.-There shall be no evidence of explosive leakage through the cone-body joint or through the liner-conduit seal.

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3.4 Propellant weight.-The weight of the propellant shall not differ by more than plus or minus 0.12 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.5 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.4.

3.5.1 Muzzle velocity.-The corrected muzzle velocity when fired at 70 degrees Fahrenheit (°F) shall not be greater than 4015 feet per second (fps) or less than 3885 fps and the standard deviation for the lot shall not be greater than 15.3 fps.

3.5.2 Functioning and penetration.-The projectile shall function and there shall be complete penetration of a 6 inch thick homogeneous armor plate when the armor plate is positioned at 60 degrees from the vertical (see 6.6).

3.5.3 Tracer functioning.-The tracer shall ignite and there shall be visible evidence of burning of the tracer composition to the 2000 meter target.

3.5.4 Safety.-There shall be no hangfire, premature burst or indication of metal parts failure in the gun bore or in flight prior to impact during any of the proving ground tests conducted.

3.6 Torque Requirements

3.6.1 Assembly torque.-Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.6.2 Disassembly torque.-Threaded joints shall withstand application of the minimum disassembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.7 Workmanship

3.7.1 The cartridge and all components shall be free of cracks or burrs; and chips, dirt, grease, rust or other

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foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The cartridge shall not be damaged in any manner during the loading operation.

3.7.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated. (Bare spots in painting, other than marking, are allowable provided that they total not more than 1/4 square inch in aggregate, and that the maximum dimension of any individual bare spot does not exceed 3/16 inch.)

3.7.3 Loading of high explosive.-The interior surface of the projectile shall be dry at the time the high explosive is poured. Solid explosive shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.7.4 The exterior of the projectile, the threads and the fuze cavity shall be cleaned to remove extraneous explosive and foreign material. If a chemical cleaner is required, no material other than acetone shall be used. Care should be taken to prevent acetone from coming into contact with the charge and becoming entrapped in the projectile. After loading, the projectile shall be clean and free of extraneous explosive.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to

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the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Quality Conformance Inspection

4.2.1 Lot Formation.-The term "inspection lot", as used in the specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions, and complies with the provisions for submission of product, as specified in

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Standard MIL-STD-105. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Cartridge cases of one type of metal and of one lot interfix number from one manufacturer, in accordance with the same drawing and drawing revision, same specification and specification revision.

b. Projectile metal parts from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision.

c. Propellant from not more than one lot.

d. Loaded primers of one lot interfix number from one manufacturer.

e. Loaded fuzes of one lot interfix number from one manufacturer.

f. Loaded tracers of one lot interfix number from one manufacturer.

g. Composition B of one lot interfix number from one manufacturer.

h. Nose lucky and insulator assemblies of one lot interfix number from one manufacturer.

i. Spike caps of one lot interfix number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100

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percent (%). Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects, using an AQL of 0.40% for each Major Defect, and an AQL of 0.65% for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor defects listed in the applicable subparagraph of this section, except where 100% inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile loading assembly, prior to loading (see dwg 8822480 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection (see 6.2)
Critical:	None defined.	
Major:	AQL 0.40 percent	
	101. Foreign matter in charge cavity	Visual
Minor:	AQL 0.65 percent	
	201. Evidence of poor workmanship (see 3.7)	Visual

4.2.2.2 Projectile loading assembly, prior to assembly of fuze shock pad (see dwg. 8822480 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:		
	1. Surface cavitation excessive (see 3.3.5)	Visual
	2. Explosive in threads, on fuze shoulder area, or on rear face of body	Visual
Major:	AQL 0.40 percent	
	101. Depth of fuze cavity incorrect	Gage (CD)
	102. Conduit damaged (after drilling operation)	Visual

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Minor: AQL 1.00 percent
201. Diameter of fuze cavity incorrect .....Gage (CD)
202. Foreign matter in fuze cavity .....Visual

4.2.2.3 Projectile loading assembly, prior to insertion of fuze (see dwg. 8822480 covering a detail of dwg 8822481).

Categories	Defects	Method of Inspection
Critical:		
	1. Fuze shock pad missing	Visual
	2. Loose explosive or chips in cavity	Visual
Major: None defined.		
Minor: None defined.		

4.2.2.4 Projectile loading assembly, prior to assembly of fuze lock plug (see dwg. 8822480 covering a detail of dwg 8822481).

Categories	Defects	Method of Inspection
Critical:		
	1. Fuze does not seat on pad	Visual/Manual
Major:	AQL 0.65 percent	
	101. Fuze terminal screw damaged or loose	Visual/Manual
	102. Wire improperly assembled to fuze, wire terminal or wire assembly damaged or loose	Visual/Manual
	103. Fuze pin does not enter chamber slot	Visual/Manual
Minor:	AQL 0.65 percent	
	201. Evidence of poor workmanship (see 3.7)	Visual

4.2.2.5 Projectile loading assembly, prior to assembly of fin and boom (see dwg. 8822480 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:		
	1. Gasket missing or damaged	Visual

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2. Fuze lock plug missing, cracked, or obviously loose .....Visual/Manual

Major: 100 percent inspection
101. Obturator inverted .....Visual

Major: AQL 0.40 percent
102. Fuze lock plug fails to withstand min. assembly torque (see 3.6 and 4.3.2)....Test (CD)

Minor: AQL 0.65 percent
201. Thread sealing compound missing from boom threads .....Visual

4.2.2.6 Projectile loading assembly, prior to assembly of tracer plug and disc assembly (see dwg. 8822480 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.65 percent	
	101. Tracer missing or assembled in an inverted position	Visual
	102. Fin and boom assembly fails to withstand min. assembly torque (see 3.6 and 4.3.2)	Test (CD)
	103. Thread sealing compound missing from threads	Visual
Minor:	AQL 0.65 percent	
	201. Evidence of poor workmanship (see 3.7).	Visual

4.2.2.7 Projectile loading assembly, prior to assembly of spike cap (see dwg. 8822480 covering a detail of 8822481).

Categories	Defects	Method of Inspection
Critical:		
	1. Washer missing	Visual
Major:	AQL 0.40 percent	
	101. Lucky assembly missing or damaged	Visual

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Minor: AQL 1.00 percent

201. Foreign material on spike threads or spike cap threads .....Visual
202. Evidence of poor workmanship (see 3.7).Visual

4.2.2.8 Projectile loading assembly (see dwg 8822480 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 1.00 percent	
101.	Gap present at any joint	Gage (CD)
102.	Tracer plug and disc assembly fails to withstand min. assembly torque (see 3.6 and 4.3.2)	Test (CD)
103.	Spike cap fails to withstand min. assembly torque (see 3.6 and 4.3.2) ...	Test (CD)
104.	Concentricity of fins with bourrelet incorrect	Gage (CD)
105.	Any components damaged	Visual
Minor:	AQL 0.65 percent	
201.	Evidence of poor workmanship (see 3.7).	Visual

4.2.2.9 Liner, prior to assembly into cartridge case (see dwg 8865255 covering a detail of dwg. 8865256).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 1.00 percent	
201.	Excessive amount of uncoated surface.	Visual
202.	Evidence of poor workmanship, including liner film torn or damaged (see 3.7)	Visual

4.2.2.10 Case, cartridge and liner assembly (see dwg. 8865256 covering a detail of dwg. 8822481).

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Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	AQL 0.40 percent	
101.	Liner installed incorrectly in that surface exposed is not that which is completely covered with film	Visual
Minor:	AQL 1.50 percent	
201.	Depth of liner from mouth of case incorrect	Gage (CD)
202.	Gap to the extent that propellant can be entrapped between liner and case	Visual
203.	Ends of cloth overlap	Visual
204.	Evidence of poor workmanship (see 3.7)	Visual

4.2.2.11 Primer, prior to assembling into cartridge case and liner assembly (see dwg. 8839485 covering a detail of dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:		
1.	One or more flash holes missing	Visual
Major: None defined.		
Minor: None defined.		

4.2.2.12 Cartridge case and liner assembly, after assembly of primer, but prior to loading of propellant (see dwg. 8822481).

Categories	Defects	Method of Inspection
Critical: None defined.		
Major:	100 percent inspection	
101.	Primer above flush	Straight-edge
Major:	AQL 0.65 percent	
102.	Primer fails to withstand disassembly torque (see 3.6 and 4.3.2)	Test (CD)
103.	Adhesive missing from mouth of case	Visual

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104. Foreign matter in case .....Visual

Minor: AQL 0.65 percent
201. Evidence of poor workmanship (see 3.7).Visual

4.2.2.13 Cartridge assembly (see dwg. 8822481).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
	101. Loading plug fails to withstand disassembly torque (see 3.6 and 4.3.2)	Test (CD)
	102. Loading plug above flush	Straight-Edge
Minor:	AQL 1.50 percent	
	201. Marking incorrect or unidentifiable including incorrect color code	Visual
	202. Adhesive on obturating band	Visual
	203. Evidence of poor workmanship (see 3.7)	Visual

4.2.2.14 Unsealed fiber container (see dwg 8800078).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:	AQL 0.40 percent	
	101. Packing component missing or out of place	Visual
Minor:	AQL 1.50 percent	
	201. Container cap cannot be removed by hand	Manual
	202. Cartridge cannot be removed by hand	Manual
	203. Glue or asphalt on cartridge	Visual

4.2.2.15 Sealed fiber container (see dwg. 8800078).

Categories	Defects	Method of Inspection
Critical:	None defined.	

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Major: AQL 0.65 percent

101.	Container cut or damaged through all impregnated layers	Visual
102.	Metal end loose or distorted	Visual/Manual
103.	Tape incomplete or badly wrinkled	Visual

Minor: AQL 2.50 percent

201.	Tear tab length under min.	Gage (CD)
202.	Contents loose	Manual
203.	Gap between cover and body of container exceeding 1/8 inch	Visual
204.	Cuts, scuffs or gouges in outer layers.	Visual
205.	Marking misleading or unidentifiable	Visual

4.2.2.16 Sealed wood packing box (see dwg. 8800077).

Categories	Defects	Method of Inspection

Critical: None defined.

Major: AQL 0.65 percent

101.	Box damaged exposing contents	Visual
102.	Board loose or broken	Visual/Manual
103.	Hardware or strapping missing, broken, or loose	Visual/Manual
104.	DOD code or other marking missing, incorrect, or unidentifiable	Visual

Minor: AQL 1.50 percent

201.	Handle missing or insecure	Visual/Manual
202.	Car seal missing, unsealed or improperly positioned	Visual
203.	Hardware improperly engaged	Visual/Manual
204.	Strapping improperly assembled	Visual/Manual

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of primer (see applicable primer specification). If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass, the check test specified in the applicable primer loading

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specification immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor facility assembling the cartridge (see 6.4).

4.2.3.2 Cavitation (see 3.3, except 3.3.5).-Cavitation defects are classified as follows:

Critical:

1. Cavitation, cracks or annular rings in Segment A in excess of that permitted by the applicable requirement.
2. Cavitation in Segment B in excess of twice that permitted by the applicable requirement.
3. Evidence of explosive leakage.

Major:

101. Cavitation, cracks or annular rings in Segment B in excess of that permitted by the applicable requirement, but not more than twice that permitted by the applicable requirement.
102. Cavitation, cracks or annular rings in Segment C in excess of that permitted by the applicable requirement.
103. Piping cavity in excess of the applicable requirement.

The sampling plan and procedure for this test for both critical and major defects shall be in accordance with MIL-STD-1235 using a CSP-2 plan having an AQL of 0.65 percent, except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing on a sample basis. When a major defect is found during the f phase of the plan, inspection shall continue at the same frequency until the homogeneity requirement is fulfilled or 100 percent inspection is required. Appropriate i and f values shall be as specified by the procuring activity. The test shall be performed using contractor designed equipment in accordance with 4.3.1.

4.2.3.3 Projectile electrical tests (see dwg. 8822480, covering a detail of dwg. 8822481).

4.2.3.3.1 Resistance between eyelet at nose end of spike and projectile body (after loading and assembly of fuze

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and fuze lock plug), Critical defect. This test shall be performed 100 percent. Any assembly which fails to meet the applicable requirement shall be classed defective and removed from the lot. Fuze circuit resistance of 1,000 to 10,000 ohms is an indication of an armed fuze and appropriate safety precautions shall be taken with projectiles with resistance in this range. The test shall be performed using contractor designed equipment.

4.2.3.3.2 Resistance between spike and spike cap (after assembly of lucky), Major defect. Sampling for this test shall be in accordance with MIL-STD-1235 using a CSP-2 plan having an AQL of 0.40 percent. Appropriate i and f values shall be as specified by the procuring activity. The test shall be performed using contractor designed equipment.

4.2.3.4 Propellant weight (see 3.4) Major defect. The propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing, and if performed manually, shall be performed by another operator. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed using contractor designed equipment.

4.2.3.5 Profile and alignment (see dwg. 8822481). Major defect. The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with drawing 7258441.

4.2.3.6 Bullet pull (see dwg. 8822481) - Major defect. The sampling plan for this test shall be in accordance with Standard MIL-STD-414. Section B, Part II, Table B-3, Code Letter F and and AQL of 1.50 percent shall be used. The test shall be performed using contractor designed equipment in accordance with 4.3.3.

4.2.3.7 Proving Ground.-The following tests shall be performed as specified in 4.3.4.

4.2.3.7.1 Muzzle velocity (see 3.5.1)-Major defect.

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4.2.3.7.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, the sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code Letter I and an AQL of 1.00 percent. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.75 exceeds the lot requirement specified in 3.5.1, the lot shall be rejected.

4.2.3.7.1.2 After three consecutive lots have met the criteria of 4.2.3.7.1.1, the sampling plan and acceptance criteria for this test shall be in accordance with standard MIL-STD-414, Section B, Part II, Table B-3, Code Letter F and and AQL of 1.00 percent. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.65 exceeds the lot requirement specified in 3.5.1, the lot shall be rejected.

4.2.3.7.2 Functioning and penetration (see 3.5.2) - Major defect.

4.2.3.7.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, twenty-five sample cartridges shall be selected from each lot for this test. If three or more sample cartridges fail to comply with the applicable requirement, the lot shall be rejected. In totaling the number of rounds failing to meet the requirements, up to two partial penetrations shall be disregarded provided that bulging or cracking of the rear face of the armor plate occurs (see 6.6).

4.2.3.7.2.2 After three consecutive lots have met the criteria of 4.2.3.7.2.1, ten sample cartridges shall be selected from each lot for this test. If two or more sample cartridges fail to comply with the applicable requirement, the lot shall be rejected. In totaling the number of rounds failing to meet the requirements, one partial penetration shall be disregarded provided that bulging or cracking of the rear face of the armor plate occurs (see 6.6).

4.2.3.7.3 Tracer functioning (see 3.5.3) - Major defect.

4.2.3.7.3.1 Beginning with the first lot produced and

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continuing until three consecutive lots have complied with the acceptance criteria of 4.2.3.7.1.1, the sample cartridges fired in the test for muzzle velocity shall be observed for tracer functioning. If three or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.7.3.2 After three consecutive lots have met the criteria of 4.2.3.7.1.1, the sample cartridges fired in the test for muzzle velocity (see 4.2.3.7.1.2) shall be observed for tracer functioning. If two or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.7.4 Target-impact dispersion(for information only). The cartridges fired for muzzle velocity in 4.2.3.7.1 shall be observed for target-impact dispersion.

4.2.3.7.5 Extreme temperature firing (for information only). Twenty cartridges shall be selected from each of the first three production lots and fired for muzzle velocity, target-impact dispersion, and max, chamber pressure information at -40 degrees F. (10 rds.) and 140 degrees F. (10 rds.).

4.2.3.7.6 Safety (see 3.5.4) - Critical defect. All cartridges fired in the proving ground tests shall be observed for premature bursts, hangfires, or break-up of metal parts. If any cartridge fails to comply with the applicable requirement the lot shall be rejected.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" heading in 4.2.2 and in 4.2.3 (see 6.2).

4.3. Test methods and procedures

4.3.1 Determination of cavities.-The sample projectiles shall be subjected to X-ray examination. In all exposures the X-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the X-ray negative shall be made to determine compliance with applicable requirements. Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453. In addition, the sensitivity of X-ray technique shall be 2.0 percent,

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and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original X-ray picture leaves doubt as to acceptability, the projectile shall be re-X-rayed at an angle 90 degrees from that of the original exposure. If, in the examination of the projectile X-ray there is any evidence of explosive within the cavity formed by the cone and spike, such as explosive adhering to the wire which passes through the cavity, the projectile shall be re-X-rayed to obtain a well defined view of the entire cavity.

4.3.1.1 Check sensitivity of X-ray technique.-The sensitivity of X-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the X-ray film. The sensitivity of X-ray technique shall be considered satisfactory when the image of the penetrameter, including the 2T hole, is sharply defined on the X-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specified gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Standard MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 2 of this specification.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used (see 6.7). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 Torque testing.-The torque shall be applied in a tightening direction for assembly torques, and in a loosening direction for disassembly torques, while holding the body for projectile torque tests, or the cartridge case

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for the primer torque test. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.3.3 Bullet pull.-The sample cartridges shall be tested for security of crimp of the cartridge case by applying the load in a axial direction. The rate of travel of the testing machine head shall be .100 to .125 inch per minute, inclusive.

4.3.4 Proving Ground.-These tests shall be conducted at a Government proving ground in accordance with the latest revision of the applicable Material Test Procedure and Acceptance Test Procedure. The sample cartridges shall be fired alternately with an equal number of calibrated reference rounds in a gun for which the cartridge is standard and observation made for conformance to all applicable requirements.

4.3.4.1 Velocity-target impact dispersion.- The ammunition fired in this test shall be conditioned at a temperature of 70 degrees F. ± 2 degrees F. in a test chamber until all propellant is stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Vertical and horizontal hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record for informational purposes only (see 6.9). Muzzle velocity shall be determined and recorded for each round.

4.3.4.2 Functioning and penetration.-The cartridge shall be fired at a steel armor plate target, 6 inches thick, complying with Specification MIL-S-13812. The target shall be placed 400 feet from the gun muzzle at an angle of obliquity of 60 degrees from the vertical. Smear cameras shall be utilized to record plate impact on the first three lots, as a minimum. Duds, partials, and complete penetrations shall be recorded (see 6.6.).

4.3.4.3 Tracer functioning.-Cartridges fired during the velocity test shall be observed for tracer functioning

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to the 2000 meter target. The distance over which the tracer burns shall be recorded to the nearest estimated 500 meters (i.e., failure, 500 meters, 1000 meters, 1500 meters, 2000 meters).

4.3.4.4 Extreme temperature firing. (Information only).- For this test ten rounds shall be temperature conditioned at -40°F + 2°F and ten rounds shall be temperature conditioned at 140°F $\mp$ 2°F until all propellant has stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Muzzle velocity, max. chamber pressure, and target impact dispersion data shall be recorded. Smear cameras at 100 ft. from the muzzle shall be utilized to record any metal parts failure.

4.3.4.5 Safety.-All rounds fired in the proving ground tests shall be observed for hangfires, premature bursts and other apparent safety hazards.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 8800078.

5.2 Packing

5.2.1 Level A. - The cartridges in containers shall be packed in accordance with dwg. 8800077.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment

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(SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection equipment Design" for guidance. The specification, paragraph, and defect from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

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6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.4 Cost of Check Test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers, or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Clean scrap.-Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in pouring and cooling operations, and scrap formed in the drilling operations.

6.6. Penetration

6.6.1 Complete Penetration.-A "complete penetration" is defined as a hole completely through the plate. The lodging of a copper slug in the completed hole is not considered cause for rejection.

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6.6.2 Partial Penetration.-A "partial penetration" is defined as a hole produced by jet action which does not extend completely through the plate.

6.7 Photographic densities.-The "American Standard for Diffuse Transmission Density" PH 2.19, published by the American National Standards Institute, 1430 Broadway, New York, N.Y., has been found satisfactory as a source of information on film densities.

6.8 Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD-147. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.9 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, when the cartridge is procured by the U. S. Army Armament Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SARPA-QA-A-A
and SARPA-AD-E-A
Dover, New Jersey 07801

6.10 Proving Ground Test

6.10.1 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

6.10.2 Test Summary: See next page.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1315-A040

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PROVING GROUND TEST SUMMARY

Test	(Note 1) Initial	Subsequent	Requirements	Sampling Paragraph	Test Procedures Paragraph
Muzzle Velocity	25	10	a. Muzzle Velocity (3.5.1)	4.2.3.7.1	4.3.4.1
			b. Tracer Functioning (3.5.3)	4.2.3.7.3	4.3.4.3
			c. Safety (3.5.4)	4.2.3.7.6	4.3.4.5
			d. Target Impact Dispersion(Info)	4.2.3.7.4	4.3.4.1
Functioning and Penetration	25	10	a. Functioning and Penetration (3.5.2)	4.2.3.7.2	4.3.4.2
			b. Safety (3.5.4)	4.2.3.7.6	4.3.4.5
Extreme Temp	20	0	a. Muzzle Velocity(Info)	4.2.3.7.5	4.3.4.4
			b. Target Impact Dispersion(Info)	4.2.3.7.5	4.3.4.4
			c. Pressure (Info)	4.2.3.7.5	4.3.4.4
			d. Safety (3.5.4)	4.2.3.7.6	4.3.4.5

NOTE 1 - Until three (3) consecutive lots are accepted.

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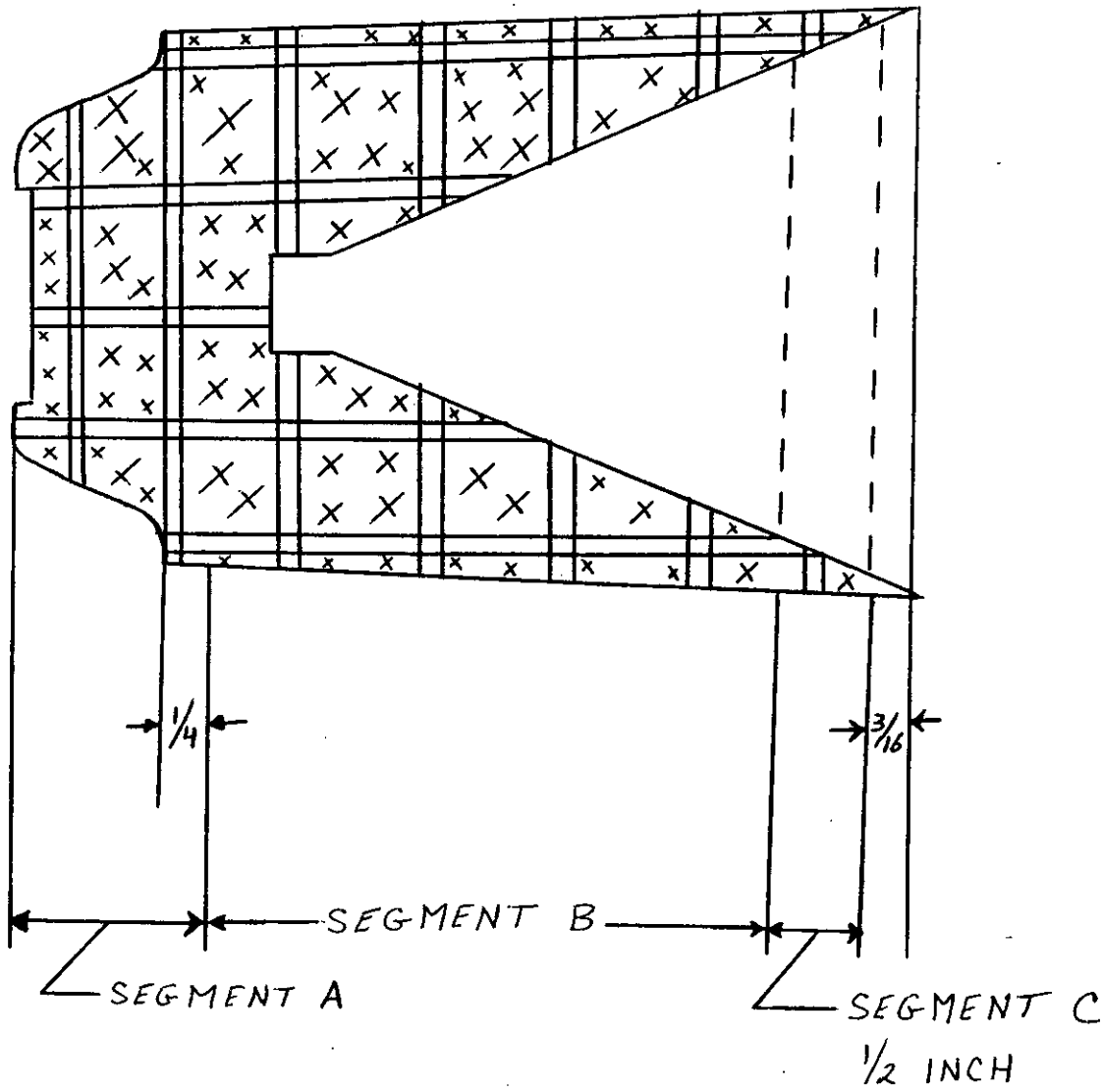
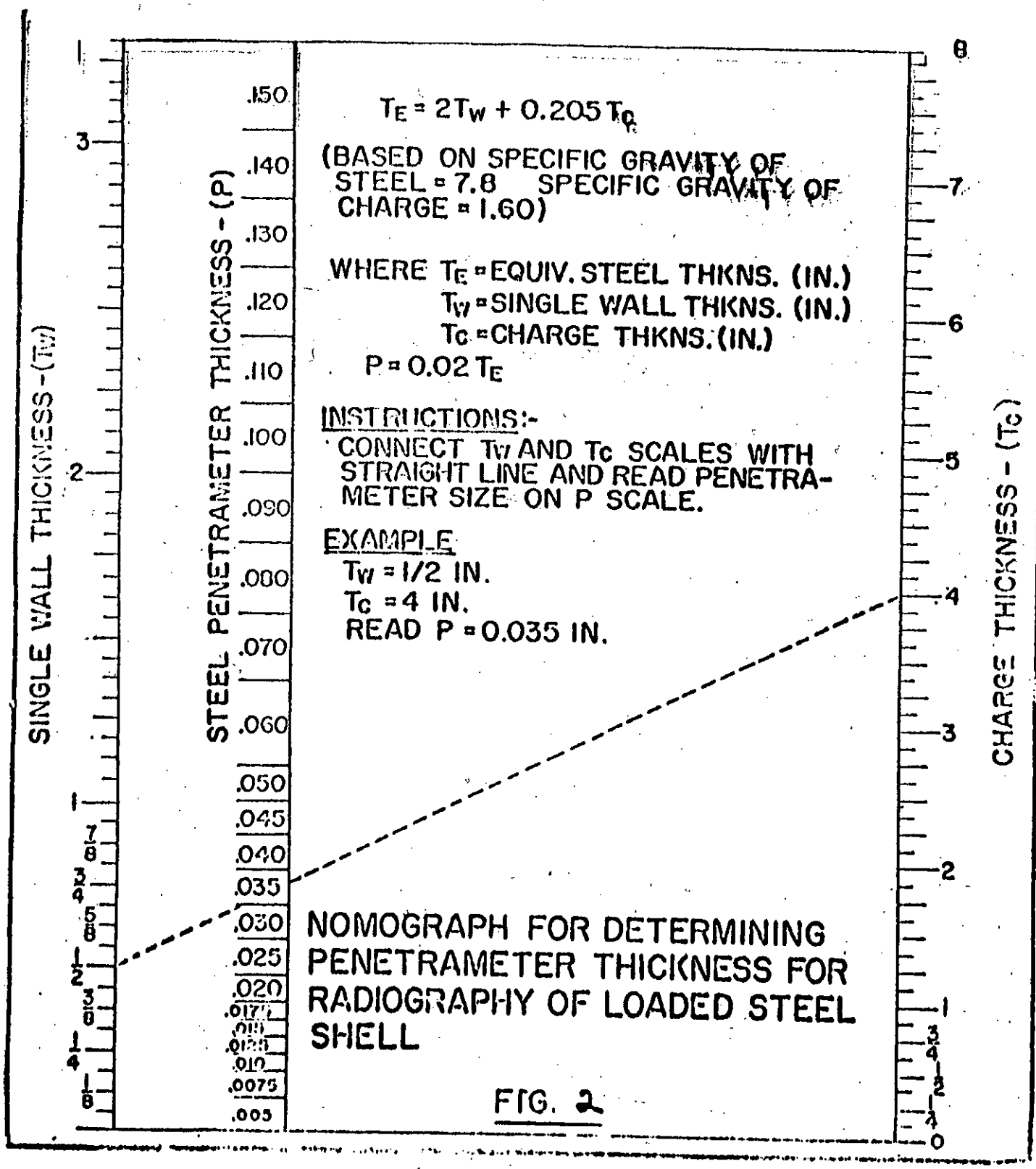


FIGURE 1

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG. 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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